

Minhal Rahman

B.Tech Graduate – Applied Electronics & Instrumentation Engineering

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PROFESSIONAL SUMMARY

Applied Electronics & Instrumentation Engineering graduate with hands-on exposure to industrial automation, PLC programming, instrumentation, sensors, actuators, HMI, SCADA, and control systems through academic projects and industrial training. Experienced in developing automation systems involving sequential control, interlocks, process monitoring, sensor-based operations, and digital-twin simulation. Familiar with CODESYS, LabVIEW, Ignition, and major PLC platforms including Siemens TIA Portal, Allen-Bradley, and ABB. Interested in automation, instrumentation, control systems, system integration, testing, field-based and project engineering roles.

EDUCATION

Government Engineering College

B.Tech, Applied Electronics & Instrumentation Engineering; CGPA: 7.24/10

Kozhikode, Kerala, India

2022 – 2026

IPCS Global

Professional Diploma in Industrial Automation (PDIA)

Calicut, Kerala, India

2026 – Present

- Coursework: PLC, SCADA, HMI, VFD, DCS, Industrial Communication.

WORK EXPERIENCE

SMEC Labs

Industrial Automation Intern

Kozhikode, Kerala, India

May – June 2025

- Gained practical exposure to PLC programming, industrial instrumentation, sensors, actuators, and control panels used in industrial automation systems.
- Worked with electrical and electronic control components, instrumentation signals, and field devices through hands-on industrial automation training.
- Developed familiarity with testing, troubleshooting, signal interfacing, process monitoring, and industrial control workflows.

PROJECTS

Smart Warehouse Automation | CODESYS, PLC Programming, OPC UA, Godot/OIP

2026

- Developed a simulated conveyor automation system involving product identification, tracking, routing, sequential operations, and automated diverter control.
- Implemented a custom Product Manager DUT with sensor-based automation, interlocks, timers, counters, I/O logic, and encoder-based position tracking.
- Connected CODESYS with a Godot/OIP 3D digital twin through OPC UA for real-time exchange of sensor, conveyor, and diverter states, and validated system behavior under different simulated conditions.

Smart Liquid Mixer (ESP32 Based) | ESP32, Sensors, Actuators, HTML, CSS, JavaScript

2025

- Developed an ESP32-based automated liquid mixing system with closed-loop gravimetric feedback, recipe-driven sequential control, and sensor-based process monitoring.
- Implemented two-stage PWM-assisted pump control to reduce overshoot near dosing setpoints, reducing average dosing error and improving repeatability by approximately 65–70%.
- Integrated load-cell/HX711 feedback, L298N motor control, sequential state-machine logic, and a web-based dashboard for real-time telemetry, recipe configuration, and process control.

PROJECTS

LabVIEW Simulation of Digital Twin <i>LabVIEW</i>	2025
• Developed a LabVIEW-based digital-twin simulation for system identification and process-control analysis, modeling the liquid mixing process and evaluating sequential batching, PWM control, and PID-based regulation. • Designed an HMI-driven simulation environment to visualize process behavior, control responses, and dosing sequences, providing a simulation-based reference for subsequent hardware implementation and validation.	

TECHNICAL SKILLS

PLC & Automation: PLC Programming, CODESYS, Allen-Bradley RSLogix 500, Siemens TIA Portal, ABB, Industrial Automation, Sequential Control, Interlocks, Timers, Counters, I/O Logic
SCADA & HMI: Ignition, InTouch, HMI Development
Instrumentation & Process Control: Industrial Instrumentation, Temperature Measurement, Level Measurement, Sensors, Actuators, Process Control, Process Monitoring, Signal Interfacing, Troubleshooting
Industrial Communication: Industrial Communication Protocols, OPC UA, TCP/IP
Embedded & Programming: ESP32, Microcontroller Programming, C, MATLAB, HTML, CSS, JavaScript
Engineering Tools: LabVIEW, Zelio Soft, Godot OIP, VS Code, Git/GitHub, Microsoft Office

LANGUAGES

Malayalam: Native Proficiency English: Professional Working Proficiency Hindi: Elementary Proficiency
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CERTIFICATIONS & WORKSHOPS

Control Engineering NPTEL, IIT Madras	2025
Siemens Mobility – Commercial Project Manager Job Simulation Forage	2026
Embedded Systems & Automotive Workshop (ENDORA’23) GEC Kozhikode	2023